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**HK IMO Training 2023-2024**  
**Problem Set 5**  
**For 4 November 2023**

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**Problem 17.** Find all function(s)  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(x^2)f(y^2) + |x|f(-xy^2) = 3|y|f(x^2y)$$

for any  $x, y \in \mathbb{R}$ .

**Problem 18.** There is a circular chessboard with 100 cells labelled  $0, 1, 2, \dots, 99$  arranged in this order. We put a chess piece in the cell with label 0. We choose three distinct positive integers  $a, b, c$  from  $1, 2, \dots, 99$  and carry out the following operations in order:

- Move the chess piece  $a$  cells forward and put a coin in the cell that it lands on. Repeat this 33 times.
- Move the chess piece  $b$  cells forward and put a coin in the cell that it lands on. Repeat this 33 times.
- Move the chess piece  $c$  cells forward and put a coin in the cell that it lands on. Repeat this 33 times.

At most how many coins can we put in the cell with label 1 at the end?

**Problem 19.** Let  $I$  be the incentre of  $\triangle ABC$ . The angle bisectors of  $\angle B$  and  $\angle C$  meet the opposite sides at  $D$  and  $E$  respectively. Let  $P$  be the point on the circumcircle  $\omega$  of  $\triangle ADE$  such that  $\angle PAI = 90^\circ$ . The line  $PI$  intersects  $\omega$  again at  $Q$ . Prove that the circumcentre of  $\triangle AIQ$  lies on the line  $BC$ .

**Problem 20.** Find all positive integers  $m$  and  $n$  such that

$$k^2m^2 + 1 \mid n^{\varphi(k)} - 1$$

for every positive integer  $k$ .